

Vipers

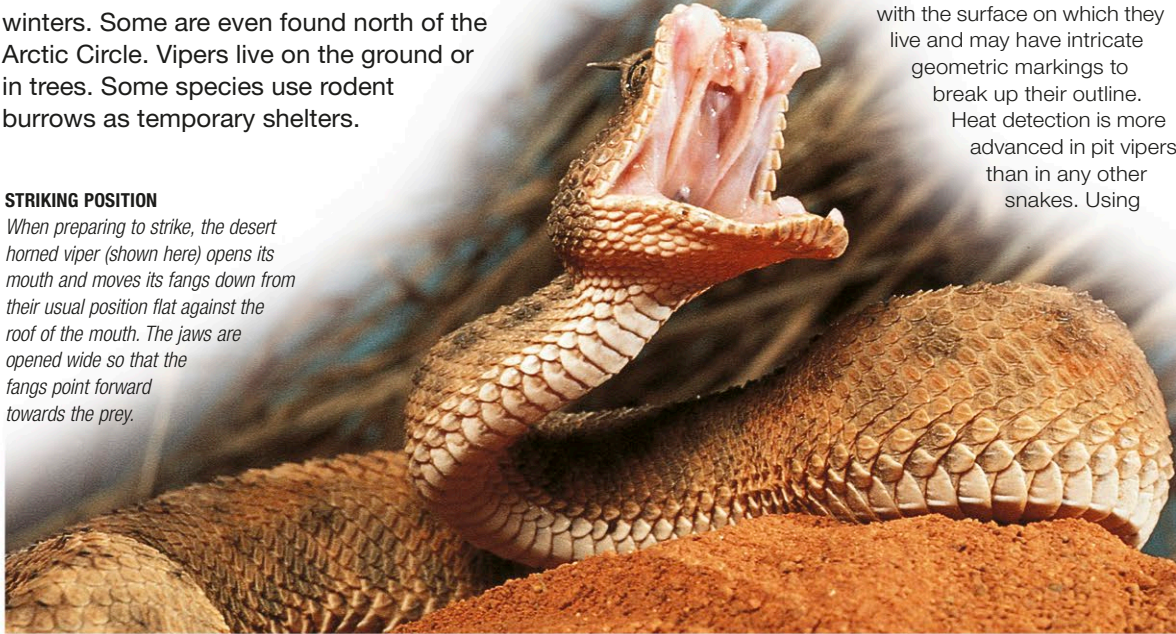
PHYLUM	Chordata
CLASS	Reptilia
ORDER	Squamata
SUBORDER	Serpentes
SUPERFAMILY	Caenophidia
FAMILY	Viperidae
SPECIES	337

Vipers are the most highly evolved of all snakes. They have long, hinged fangs, and some (the pit vipers) also have a pair of heat-sensitive pits between their eyes and nostrils. Vipers are found in a wide range of climates, and compared to other snakes are better equipped to deal with cold conditions. Many live at high elevations or in desert areas with cold

winters. Some are even found north of the Arctic Circle. Vipers live on the ground or in trees. Some species use rodent burrows as temporary shelters.

STRIKING POSITION

When preparing to strike, the desert horned viper (shown here) opens its mouth and moves its fangs down from their usual position flat against the roof of the mouth. The jaws are opened wide so that the fangs point forward towards the prey.



Anatomy

Most vipers are short and stocky with wide, triangular heads and rough, keeled scales. The head has long fangs, which are folded against the roof of the mouth when not in use, and large venom glands. Vipers are generally sluggish snakes and rely on camouflage to go undetected: they are often colored to blend in with the surface on which they live and may have intricate geometric markings to break up their outline. Heat detection is more advanced in pit vipers than in any other snakes. Using

their heat-sensitive pits, they can not only detect the presence of prey but also gauge its distance and the direction in which it lies. Rattlesnakes, which are a type of pit viper, have a unique warning device in the form of a rattle at the end of the tail.

Venom

Most vipers ambush their prey, some using the brightly colored tip of their tail as a lure. They lunge open-mouthed and stab the victim using their long fangs, penetrating fur and feathers, to reach the vital organs. The venom varies from species to species, but often contains proteins that break down blood cells and cause internal hemorrhaging. The venom being relatively slow acting, vipers deliver large amounts of it, unlike cobras. For the prey, death may not be instantaneous.

Reproduction

Although some vipers lay eggs, most give birth to live young. Females retain the developing eggs in their body and bask during the day to accelerate the embryos' development. Many species breed only every 2, or even 3 years, using the nonbreeding years to feed and recover their body weight.

REPTILES

Atheris mabuensis

Mount Mabu forest viper



Length 15 in (38 cm) max
Breeding Viviparous
Habit Terrestrial
Status Not evaluated

Location E. Africa



This is the mostly southerly—and smallest—of *Atheris* African forest vipers; it was described as a new species in 2009 and is known only from the midaltitude rain forest of Mount Mabu and Mount Namuli in northern Mozambique. Its highly restricted distribution makes it at risk from habitat destruction. Adults resemble the juveniles of larger allied species and, unlike most members of its genus, are ground dwelling. It may be a specialized hunter—taking small frogs and geckos among leaf litter.



Bitis arietans

Puff adder

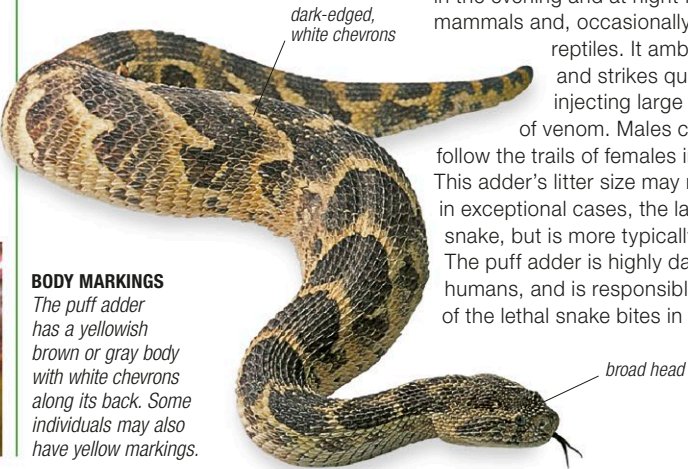


Length 3¼ ft (1 m), max 6¼ ft (1.9 m)
Breeding Viviparous
Habit Terrestrial
Status Not evaluated

Location Africa (south of Sahara and Morocco), Saudi Arabia



The puff adder is a thick-bodied snake with a wide, flattened head and a rounded snout. The male is smaller and often more brightly colored than the female. Sluggish and slow moving, larger individuals crawl in a straight line,



BODY MARKINGS
The puff adder has a yellowish brown or gray body with white chevrons along its back. Some individuals may also have yellow markings.

DEFENSIVE BEHAVIOR



Highly venomous, the puff adder is ferocious when disturbed, and defends itself by inflating its body and making a long, low hissing sound. If further provoked, it may strike, delivering a fatal bite.

like a caterpillar. The puff adder hunts in the evening and at night for small mammals and, occasionally, birds and reptiles. It ambushes prey and strikes quickly, injecting large amounts of venom. Males combat and follow the trails of females in the spring. This adder's litter size may reach 154 in exceptional cases, the largest of any snake, but is more typically 20–40. The puff adder is highly dangerous to humans, and is responsible for most of the lethal snake bites in Africa.

Bitis caudalis

Horned adder



Length 12–20 in (30–50 cm)
Breeding Viviparous
Habit Terrestrial
Status Not evaluated

Location Southern Africa



The horned adder is gray, brown, reddish, or orange, with a single horn over each eye. Its flattened body and rough scales help it shuffle down into loose sand in order to escape the heat and to hide. When crossing loose sand, it may move in a sideways looping motion, known as sidewinding. It hunts mainly in the evening—prey may be attracted to its twitching tail.

